

Cohort Analysis Report with Visual Insights

Excelerate AI-Powered Data Analysis Remote Internship — Week 2 Deliverable

Team: RIT 1307 AI Data Team 1

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1. Dataset Overview

This report brings together the data cleaning, feature engineering, exploratory analysis, and visualization work completed by the team for Week 2. The dataset (RIT_CLEANED_DATA_SHEET) tracks 8,558 learner records from January 2023 to March 2024, following learners across internships, courses, competitions, events, and short engagements — from sign-up, through acceptance, to starting an opportunity and, in some cases, dropping out.

2. Data Cleaning and Feature Engineering

2.1 Cleaning Steps Performed

- Date/Time standardization: text-based date fields were converted to proper Excel Date/Time format. 'NA' placeholders were blanked before conversion to avoid errors while preserving missingness, and conversions were validated with YEAR() and ISNUMBER().
- A Learner SignUp Date field was derived from the Learner SignUp DateTime timestamp for temporal analysis, while the original timestamp was preserved.
- Entry Created At was found to be constant across all records, so a separate Entry Time feature was derived to retain the time-of-day component.
- Categorical standardization: TRIM() and PROPER() were applied to remove stray whitespace and normalize capitalization; inconsistent institution and country name variants were consolidated.
- Data quality validation: a set of records where Opportunity Start Date fell after Opportunity End Date were flagged as likely data-entry errors. These were retained (not deleted) since the correct dates could not be independently verified, and are marked for correction at the source.

2.2 Derived Features

Feature	Source	Method	Purpose
Learner SignUp Date	Learner SignUp DateTime	=INT(DateTime)	Registration trend analysis
Registration Year	Learner SignUp Date	=YEAR(...)	Year-over-year registration patterns
Registration Month	Learner SignUp Date	=TEXT(...,"mmm")	Monthly/seasonal registration trends
Entry Time	Entry Created At	=MOD(DateTime,1)	Time-of-day analysis
Age	Date of Birth	=DATEDIF(DOB,TODAY(),"Y")	Demographic analysis
Accepted Applications	Status Description	=IF(Status="Accepted",1,0)	Application outcome / modeling target support
Opportunity Duration (Days)	Start/End Date	IFS() median-duration lookup by type	Compare opportunity length
Dropped Out	Status Description	=IF(Status="Dropped Out",1,0)	Dropout rate measurement

3. Exploratory Data Analysis with Visual Insights

3.1 Learner Demographics

The learner population is predominantly young adults, concentrated in the 20–30 age range. The 25–30 bracket is the single largest group for both genders, followed by 20–25. Male participants slightly outnumber female participants overall, though the two genders follow a very similar age distribution.

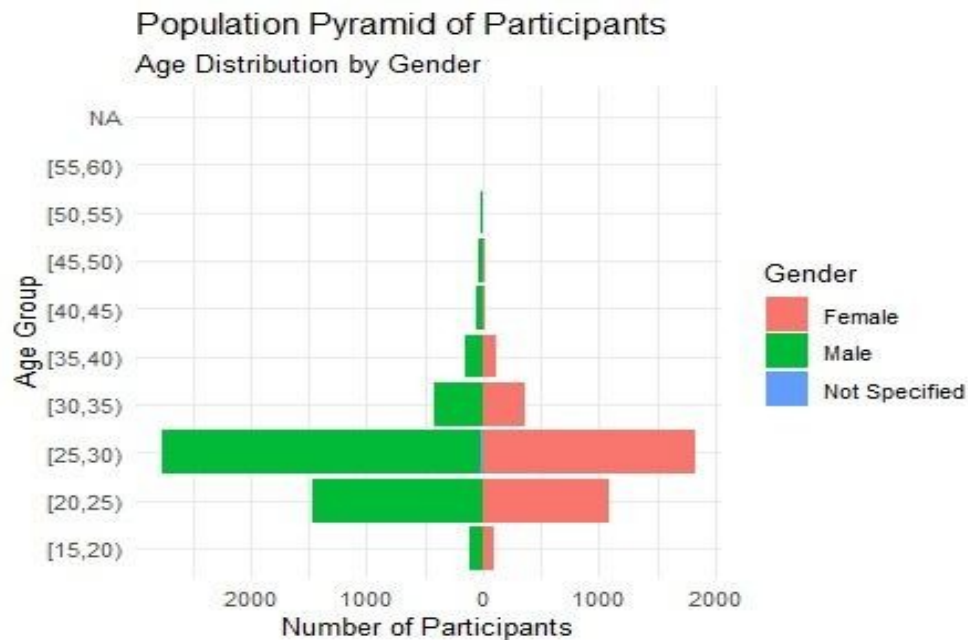


Figure 1. Age distribution by gender (population pyramid).

A boxplot of learner age confirms the dataset is fairly consistent — centered in the mid-20s — but surfaces outliers at both ends of the range (including an age-7 and a near-60 record) that need to be checked before modeling.

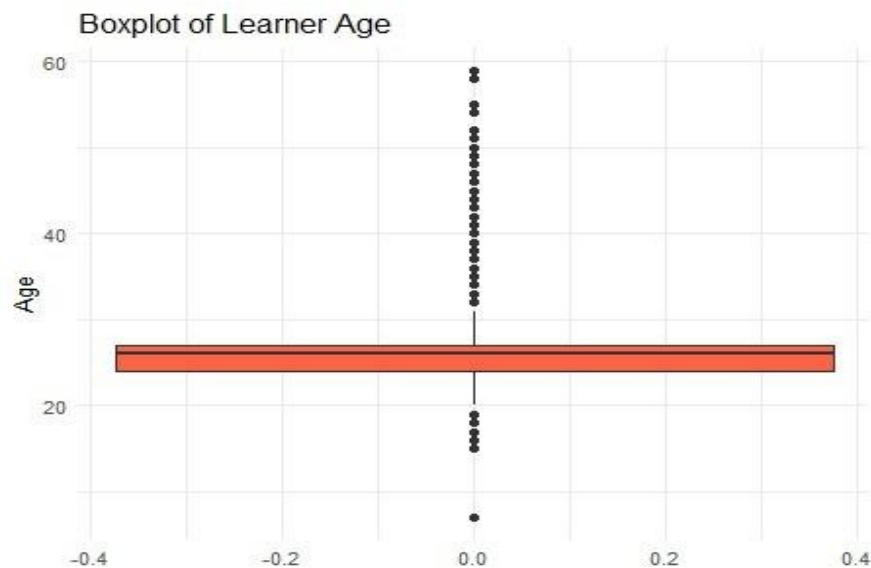


Figure 2. Boxplot of learner age, showing outliers above and below the typical range.

3.2 Registration Trends and Institutional Reach

Sign-ups are not steady through the year — they cluster around academic term breaks. January and August stand out as the two clearest peaks (consistent with major intake/recruitment windows), with a quieter stretch from September to November. The apparent drop in March 2024 is a reporting artifact: the dataset ends mid-month rather than reflecting an actual decline.

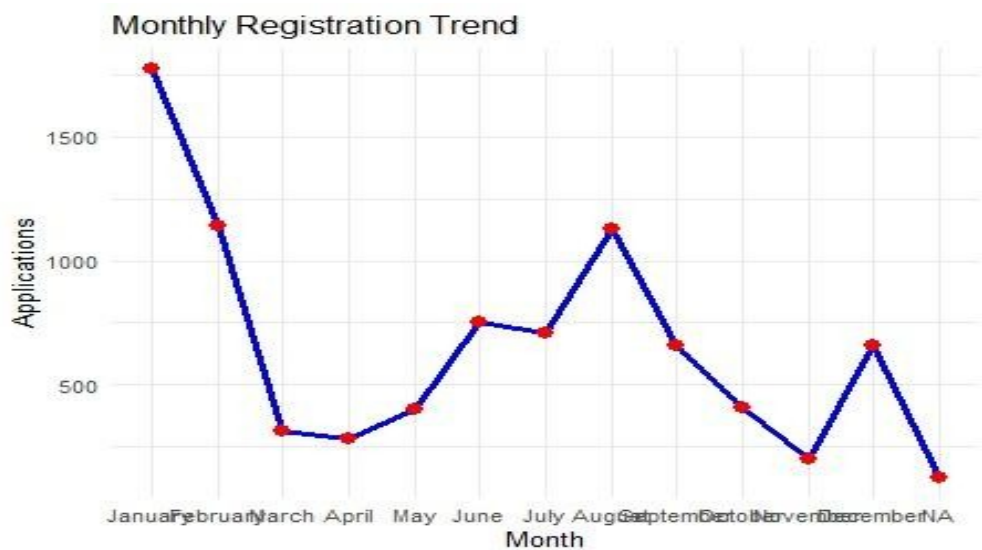


Figure 3. Monthly registration trend.

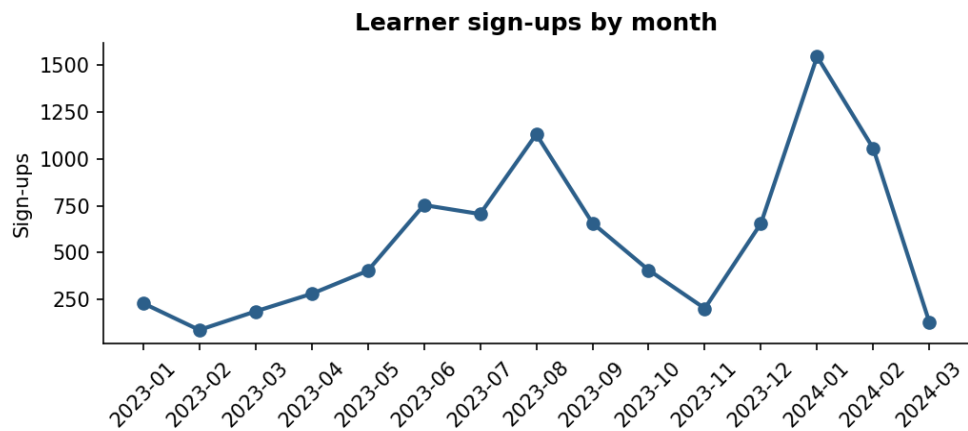


Figure 4. Monthly learner sign-ups, Jan 2023–Mar 2024 (year-specific view).

Saint Louis University is by far the dominant source institution in the dataset, contributing substantially more learners than any other school. Representation from the remaining 14 institutions in the top 15 is comparatively low, pointing to an uneven institutional spread that outreach efforts could help balance.

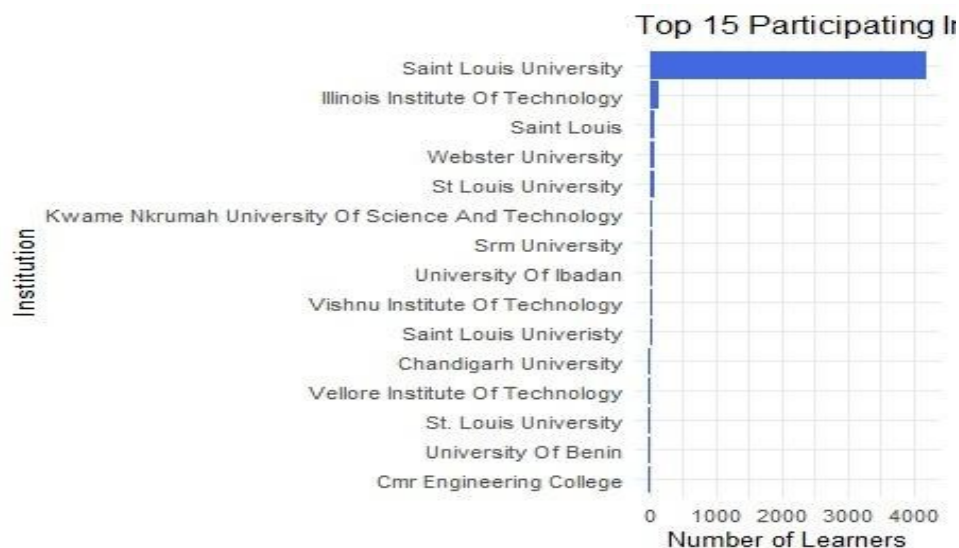


Figure 5. Top 15 participating institutions by learner count.

3.3 Where Participation Breaks Down

Internships make up 63% of all records (5,421 of 8,558). Their headline dropout rate of 11.3% looks unremarkable next to Courses, Events, and Competitions (all under 1%) — but that number is misleading, because most internship records never had a real start date logged, meaning those learners are still pending or were rejected before anything started rather than having genuinely dropped out.

Narrowing to the 1,097 internship records that have both a start and end date on file (mostly short, 30–90 day internships) reveals a much sharper pattern:

Program	Participants	Dropout rate
Digital Marketing	169	75%
Innovation & Entrepreneurship	127	73%
Data Visualization	306	71%
Project Management	255	67%
Data Visualization Associate	97	0%
Business Consulting	60	0%
Digital Strategy Virtual Internship	53	0%

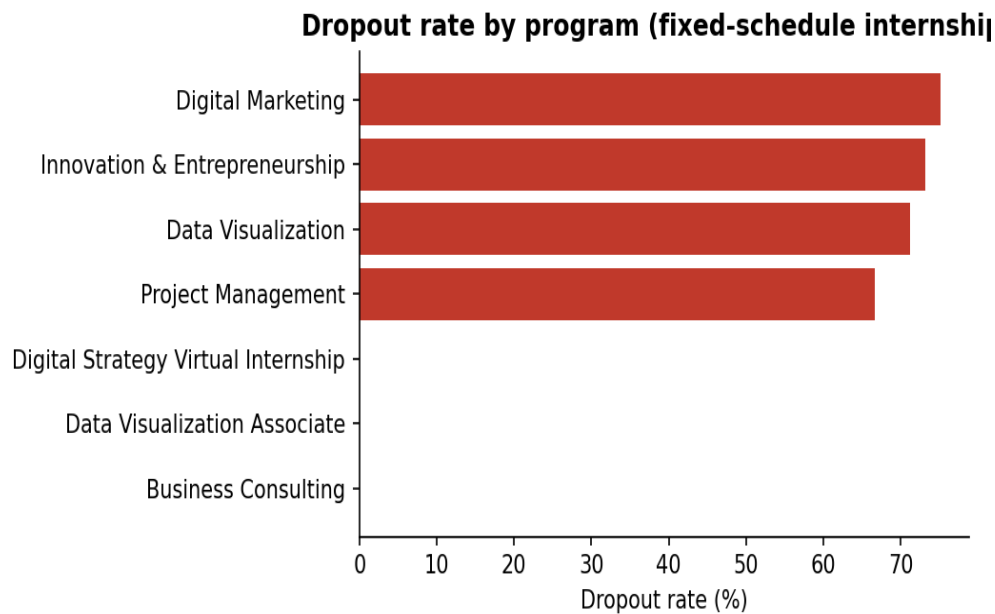


Figure 6. Dropout rate among fixed-schedule internship programs.

Just four programs — Digital Marketing, Innovation & Entrepreneurship, Data Visualization, and Project Management — account for 608 of the 617 total dropouts across the entire dataset (99%), with rates between 67% and 75%. Three comparable programs have zero dropouts. This is a program-design pattern, not a demographic one.

4. Interpretation of Findings

- Dropout risk sits almost entirely inside four specific internship programs, at rates six to ten times the platform average, while similar programs run at zero. This is the single most actionable finding in the dataset — it points to something specific about onboarding, workload, or mentor support in those four programs, worth exploring directly with participants or program staff.
- Age and gender are weak predictors of dropout. Dropout by gender is nearly identical (7.1% men vs 7.4% women). By age, 19–22 drop out least (4.4%), while 23–26 and 31–40 are highest (~7.8–8%) — but this gap is small and likely reflects which programs each age group tends to join, not age itself.
- Dropout by country is fairly even among the largest groups (US, India, Nigeria: ~11–12%). Pakistan (14%) and especially Ethiopia (29%) run higher, but both are small samples (77 and 21 people) and should be treated as a watch-item, not a confirmed finding.
- Registration is strongly seasonal, clustering around academic term breaks (January and August), which has direct implications for staffing and support-resource planning.
- Saint Louis University dominates institutional representation — a strength for that partnership, but a gap in broader institutional diversity that recruitment could address.

5. Data Quality Issues and Resolutions

Deeper EDA on the cleaned file surfaced issues that went beyond the initial cleaning pass. Each has since been addressed by the team:

- **RESOLVED** — Opportunity Duration previously returned #VALUE! for 4,528 records (53% of the dataset). The team rebuilt the calculation as a median-per-category lookup, and this has been verified directly against the current dataset: a COUNTIF check across the full Duration column returned zero #VALUE! errors. Where a genuine start date is missing, Duration now correctly shows "NA" rather than an error.
- **RESOLVED** — Missing Accepted Applications and Dropped Out values have been imputed with NaT, so the fields now behave correctly in downstream analysis and modeling.

- RESOLVED — Age outliers addressed: the age-7 record (application status: Withdraw) was removed. Of the two ~58-59 outliers, only the one with a Rejected status was removed; the remaining records in that age range are legitimate Team Allocated applicants and were correctly retained rather than dropped indiscriminately.
- STILL OPEN — 73 records show a negative Opportunity Duration (end date before start date), most likely swapped start/end dates during entry. Not yet confirmed as fixed.
- STILL OPEN — Entry Created At is not usable for timing analysis: every value falls within the same two-minute window on 2024-11-03, confirming it is an export timestamp rather than a real activity date. No fix needed if this field is simply excluded from modeling.
- STILL OPEN — 3,281 records (39%) show an Apply Date earlier than the recorded Sign-Up date. This has not yet been checked with whoever manages the source system.

6. Readiness for Predictive Modeling (Week 3)

The dataset's most significant blocker — the broken Duration field — is now confirmed fixed, along with missing-value handling and age outlier cleanup. Two smaller items remain open but do not block Week 3 from starting:

- Confirm or dismiss the 73 negative-duration records.
- Decide how to treat the Sign-Up vs. Apply Date mismatch (39% of records) before using either field as a model input.

With the major fixes in place, the dataset and features below are well-positioned for Week 3:

- Target variable: Dropped Out — but modeled per-program or with Opportunity/Program as a feature, since dropout is concentrated in four specific programs rather than driven by demographics.
- Useful predictors: Accepted Applications, Registration Month (captures seasonality), the now-corrected Opportunity Duration, Age (post-outlier-cleanup), and Country (with small-sample caution for Pakistan/Ethiopia).
- Weak predictors, safe to deprioritize: Gender and Age alone show little independent signal on dropout.

7. Team Charter

Team Charter: https://drive.google.com/file/d/1PwXa-JZ_fHiqgH1fYdyAgdo_Vjq5dU-O/view?usp=drivesdk